Slippage Desk

The options agent that measures its own execution.

Alpaca AI Trading Agents Hackathon · paper account PA343VC6LL3T

Every agent here optimises *what* to trade.

None of them measure whether they can actually get the price their edge assumed.

A win banks	+$32.43
The stop costs	−$81.09
Breakeven win rate	71.4%
Delta-implied OTM rate	76%

The entire edge is 4.6 percentage points, worth $5.19 a contract. Crossing the bid/ask cost $1.78 of that, 34% of the edge. Every figure derived from config and from the 23 contracts actually filled.

I read all 47 submissions before writing a line of this pitch.

Eleven describe the same architecture:

`"LLM proposes, deterministic gates authorise."`

That is the baseline now, not a differentiator. This project has it too. It is not the claim.

The claim is that execution quality is measurable in four days, and strategy edge is not.

Score every fill against the mid it was priced at.

- → Capture ratio kept per `(underlying, tenor, delta band, time of day)`
- → Shrunk toward 1.0, so one lucky fill cannot make a bucket look expert
- → Buckets that surrender too much credit are **refused**
- → The same memory sets how far to cross on the next order

Buckets that rarely fill lean toward crossing. Buckets that fill readily hold out for mid.
The agent closes a loop on its own execution rather than just logging it.

The model has strictly negative authority.

It may shrink a trade or veto it. Nothing else.

- → Fifteen deterministic gates run first; sizing is a fixed formula
- → The clamp **multiplies**, never assigns. A model returning 5.0 is treated as 1.0
- → It cannot pick a strike, change an expiry, or widen risk
- → Timeout, refusal, malformed JSON, NaN: every failure path is a veto

The field splits here. Of the 47 other submissions, only six say what happens when their model fails at all. This one refuses. Failure degrades to trading less, never to trading unsupervised.

Early assignment.

A short call goes in-the-money. An ex-dividend falls before expiry. Capturing the dividend beats the remaining extrinsic, so the holder exercises early.

You are assigned 100 shares per contract and carrying an overnight gap you never agreed to. The spread's defined risk was defined against expiry, not against being assigned three days before it.

Forty-seven other submissions. Two mention assignment at all.

Three surfaces, deliberately separate jobs.

SURFACE	JOB
`CLI`	Execution and book verification. Nine endpoints.
`Python SDK`	Chains with greeks and IV in one call.
`MCP server`	Read-only research for the advisor.

Execution never routes through MCP, and that is enforced rather than intended: the server launches without the `trading` toolset, so no order tool exists to call. A stdio subprocess must never sit between the agent and its stops.

`15 / 15 surfaces responding, live on the dashboard.`

Not a screenshot. A reproducible artefact.

CONSIDERED	CLEARED GATES	CREDIT CAPTURED	LOST TO EXECUTION
693	428	97.9%	$41
candidates evaluated	passed all fifteen	broker-verified	the number nobody else has

`data/proof.json` carries every fill, the mid it was priced against, the credit received, per-bucket capture, every bucket refused, and a sha256 over the payload.

Sourced from Alpaca's own activity log. Anyone holding the account ID can reproduce it.

What we claim, and what we don't.

We do not claim the strategy is profitable. Four sessions cannot establish that for anyone in this hackathon, and any entrant claiming otherwise is reading noise.

We claim the agent measured its own execution quality and acted on it, and we ship the evidence.

> *"Spot sits only 0.8% below the 721 short strike after a 1.2% down day, and the 0.75 credit on a 5-wide spread does not pay for a 2-DTE gamma gap."*
>
> The advisor, declining a trade the gates had approved.

Slippage Desk

Most agents optimise what to trade.
This one learns whether it can get the price.

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github.com/priyanshshahh/slippage-desk
dashboard-theta-lemon-36.vercel.app
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```

Built with Claude Code.